

Splunk_Security_Project

ABSTRACT

This project focuses on implementing a centralized log analysis and monitoring solution using Splunk Enterprise. The objective is to collect, index, and analyze log data from diverse sources including Kali Linux and Windows systems, enabling real-time security monitoring and operational insights. By configuring Splunk Universal Forwarders on the source systems, log data is forwarded securely to a Splunk server hosted on a Windows platform. The project covers setting up data ingestion pipelines, defining indexes and sourcetypes, creating effective dashboards and alerts for threat detection, and optimizing search queries for quick incident response. This solution aims to enhance visibility into security events, streamline log management, and support proactive threat hunting in a heterogeneous IT environment.

Introduction

Splunk is a powerful software platform designed for searching, monitoring, and analyzing machine-generated data from various sources such as servers, applications, network devices, and endpoints. It collects and indexes data in real-time, enabling users to efficiently troubleshoot system issues, monitor IT infrastructure, detect security threats, and gain valuable operational insights. Splunk's advanced search language (SPL), customizable dashboards, and alerting capabilities make it an indispensable tool for IT operations and cybersecurity professionals.

In this project, Splunk is used to centralize and analyze logs generated from both Kali Linux and Windows environments. By deploying Splunk Universal Forwarders on Kali Linux, logs such as system logs, authentication logs, and boot logs are forwarded securely to a Splunk Enterprise instance running on Windows. This setup allows real-time log ingestion, indexing, and detailed analysis, empowering the detection of security incidents, monitoring system health, and supporting proactive response strategies. The project demonstrates how Splunk facilitates comprehensive log management and threat detection in a mixed operating system environment, improving overall situational awareness and operational efficiency.

Hardware and Software Used :

Hardware:

- Personal computer or server hosting Splunk Enterprise (Windows OS)
- Kali Linux machine (physical or virtual) acting as log source and forwarder
- Network infrastructure (router/switch) for connectivity between Kali and Splunk server

Software:

- Splunk Enterprise (Windows version) — main log analysis and visualization platform
- Splunk Universal Forwarder (Linux version) — installed on Kali Linux to forward logs
- Kali Linux OS — source of logs such as system logs, authentication logs, and boot logs
- Windows OS — host for Splunk Enterprise deployment
- Supporting tools and utilities:
 - Linux shell (bash) for scripting log aggregation
 - Network and firewall configuration tools to open necessary ports (e.g., 9997) for Splunk communication

This combination enables centralized log collection, real-time indexing, search, and monitoring across diverse systems.

Environment Setup

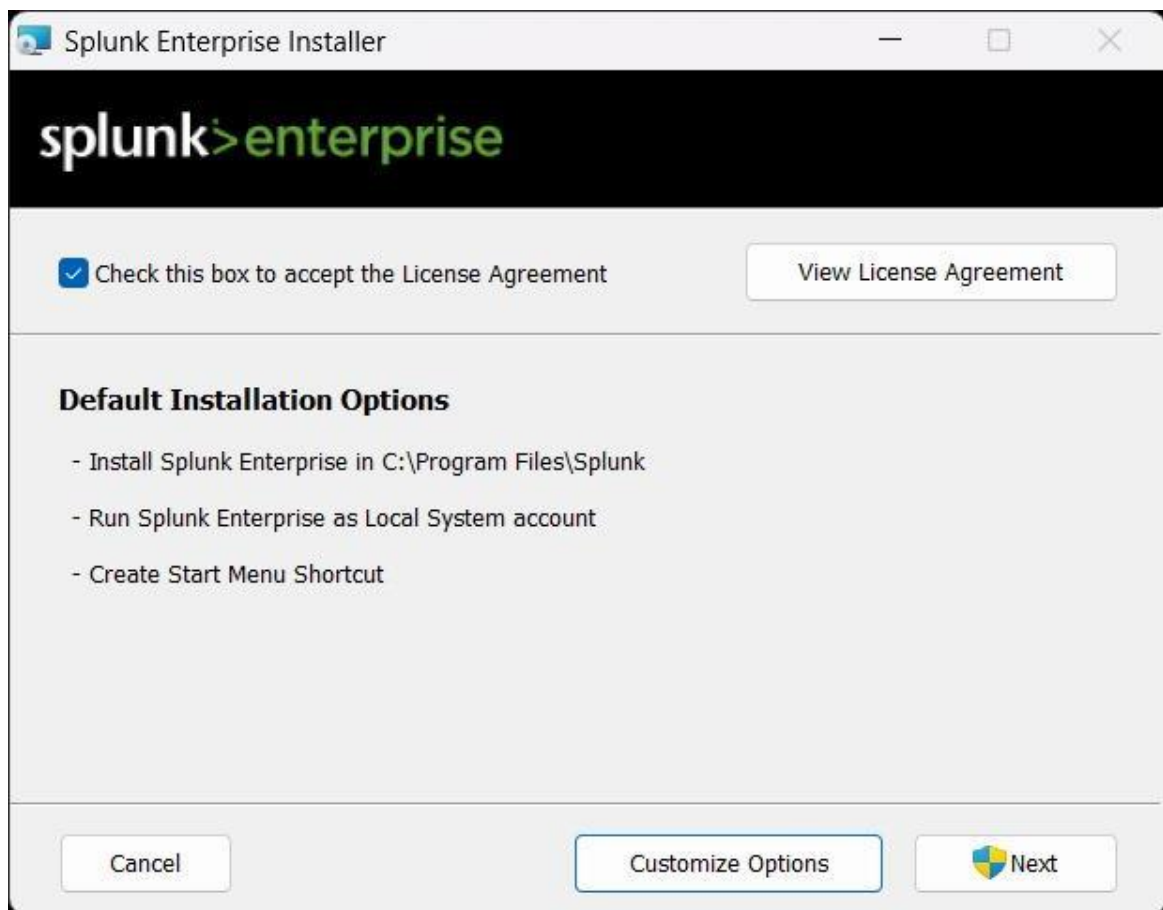
Splunk Installation Details :

- Splunk Enterprise Version: 10.0.1
- Build/Release Identifier: c486717c322b (internal build tag)
- Platform: Windows, 64-bit architecture

Splunk Installation Steps :

Download the Splunk Enterprise installer for Windows (e.g., splunk-10.0.1-c486717c322b-windows-x64.exe) from Splunk's official website.

Step 1: Accept the licence agreement by checking the box and click next.



Step 2: Create a username and password for the Splunk Enterprise and click next.

Splunk Enterprise Setup

splunk>enterprise

Create credentials for the administrator account. The password must contain, at a minimum, 8 printable ASCII characters.

Username:

Password:

Confirm password:

Cancel Back Next

Step 3: Click install and wait for the installation to complete.

Splunk Enterprise Setup

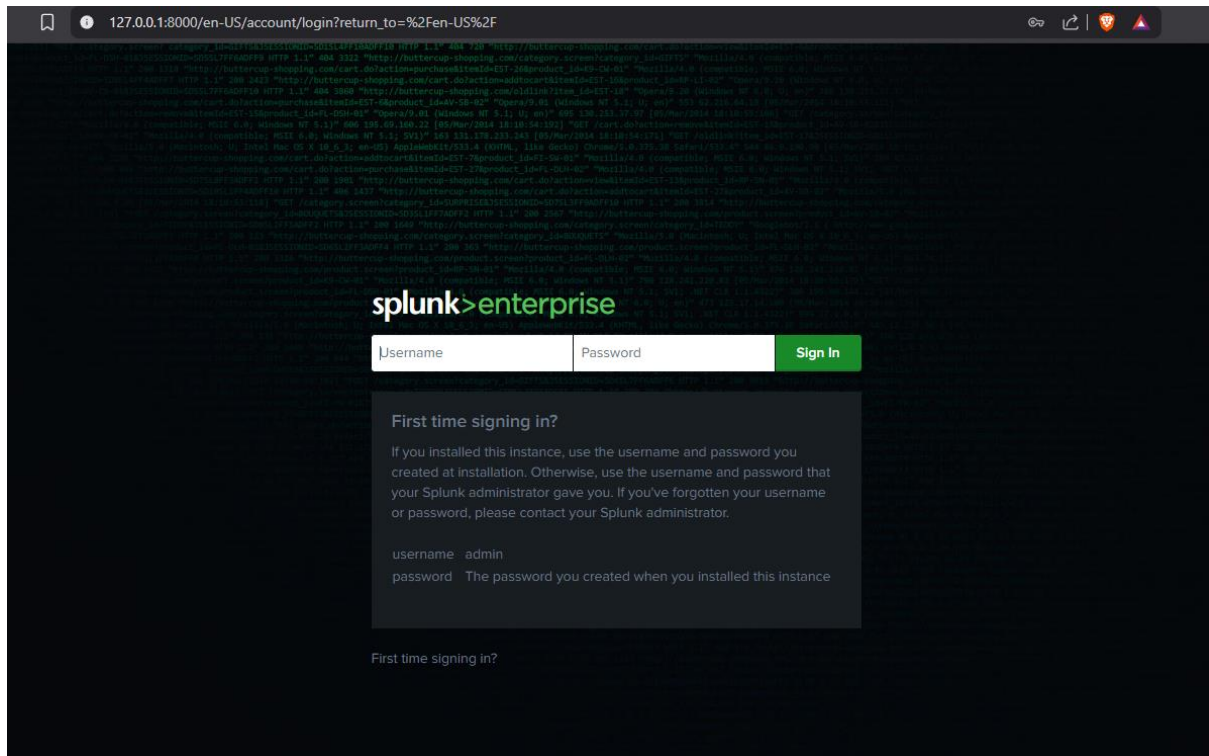
splunk>enterprise

Please wait while the Setup Wizard installs Splunk Enterprise.

Status:

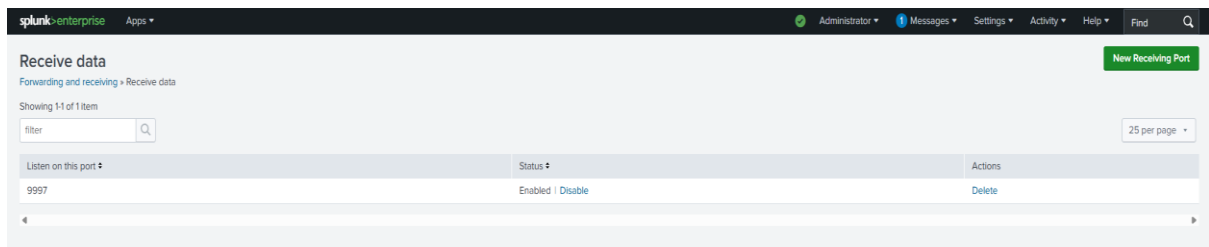
Back Next

Step 4 : Once installed, start Splunk service or open Splunk Web via <http://localhost:8000>

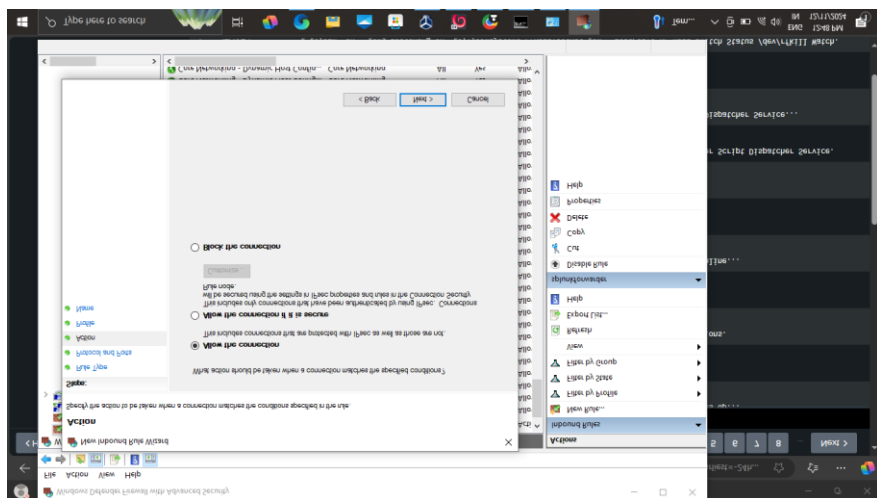
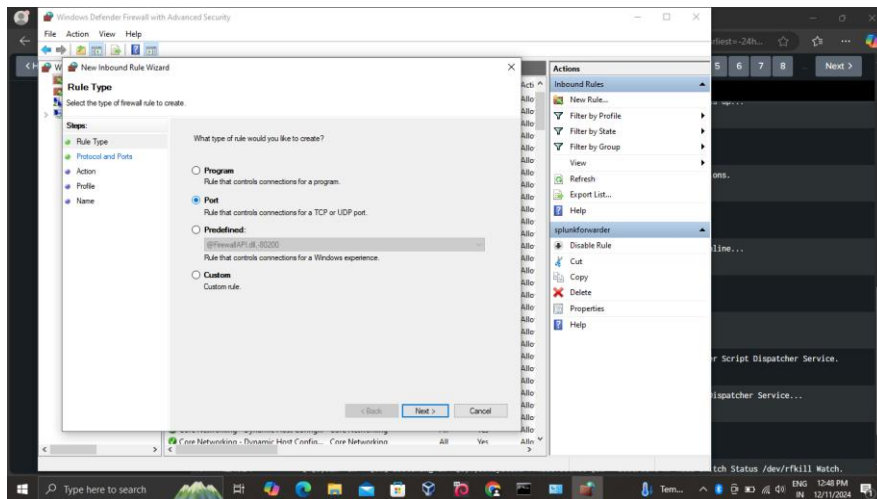
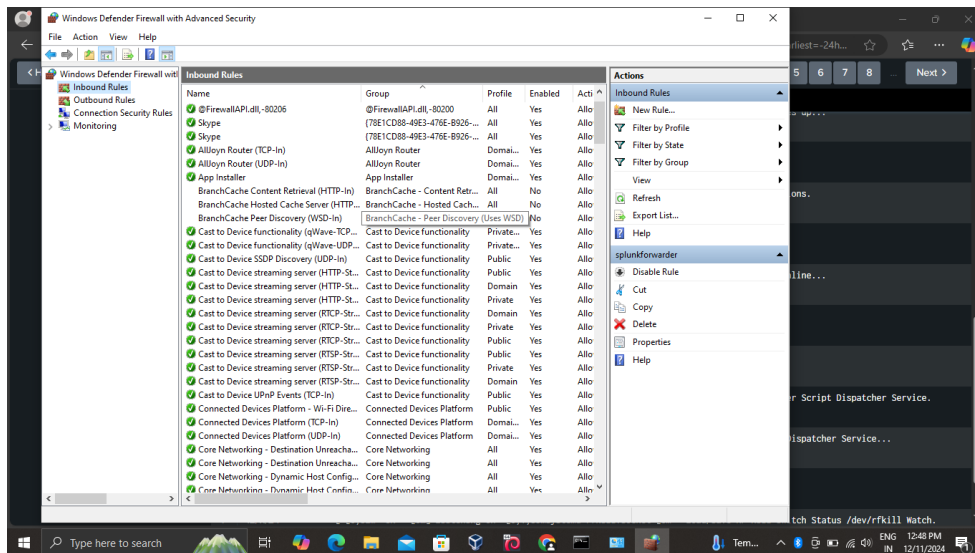


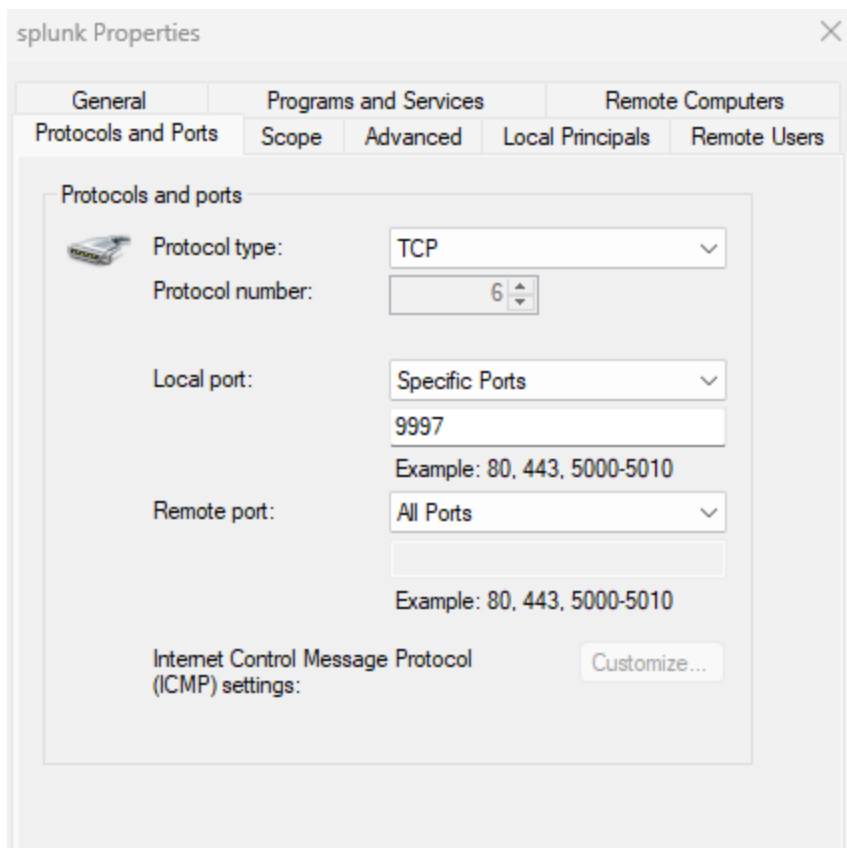
Step 5 : Configure receiving port for forwarders:

In Splunk Web, go to Settings → Forwarding and receiving → Configure receiving → Enable port (default 9997).



Step 6 : Open firewall port 9997 inbound on Windows server for forwarder connections.





Splunk Universal Forwarder Installation on Kali Linux

Step 1 : Download Splunk Universal Forwarder .deb package from Splunk's downloads page for Linux 64-bit.

```
(root@kali)~[~]
# cd /tmp
(root@kali)~/tmp
# wget -O splunkforwarder-9.3.2-d8bb32809498-linux-2.6-amd64.deb "https://download.splunk.com/products/universalforwarder/releases/9.3.2/linux/splunkforwarder-9.3.2-d8bb32809498-linux-2.6-amd64.deb"
--2024-12-06 20:19:32-- https://download.splunk.com/products/universalforwarder/releases/9.3.2/linux/splunkforwarder-9.3.2-d8bb32809498-linux-2.6-amd64.deb
Resolving download.splunk.com (download.splunk.com)... 65.8.76.39, 65.8.76.19, 65.8.76.33, ...
Connecting to download.splunk.com (download.splunk.com)|65.8.76.39|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 35707044 (34M) [application/x-debian-package]
Saving to: 'splunkforwarder-9.3.2-d8bb32809498-linux-2.6-amd64.deb'

splunkforwarder-9.3.2-d8bb32809498-linux- 100%[=====] 34.05M 12.0MB/s in 2.8s
2024-12-06 20:19:36 (12.0 MB/s) - 'splunkforwarder-9.3.2-d8bb32809498-linux-2.6-amd64.deb' saved [35707044/35707044]
```

Step 2 : Install the package on Kali:

`sudo dpkg -i splunkforwarder-9.3.2-d8bb32809498-linux-2.6- amd64.deb`

```

(root@kali)-[/tmp]
# ls
Temp-8b526ad7-12aa-41e7-9247-612a24ec3338
splunkforwarder-9.3.2-d8bb32809498-linux-2.6-amd64.deb
ssh-FC0vzVVL6W5d
systemd-private-f8860a267e3349dda35a159b88f75d4d-ModemManager.service-jK7Vw5
systemd-private-f8860a267e3349dda35a159b88f75d4d-colord.service-SpH0ux
systemd-private-f8860a267e3349dda35a159b88f75d4d-fwupd.service-KgbUZW
systemd-private-f8860a267e3349dda35a159b88f75d4d-haveged.service-PKp0rD
systemd-private-f8860a267e3349dda35a159b88f75d4d-palokit.service-cWnbIp
systemd-private-f8860a267e3349dda35a159b88f75d4d-power-profiles-daemon.service-QtUdQn
systemd-private-f8860a267e3349dda35a159b88f75d4d-systemd-logind.service-G3i0v2
systemd-private-f8860a267e3349dda35a159b88f75d4d-upower.service-QDEPjo

(root@kali)-[/tmp]
# dpkg -i splunkforwarder-9.3.2-d8bb32809498-linux-2.6-amd64.deb
Selecting previously unselected package splunkforwarder.
(Reading database ... 393193 files and directories currently installed.)
Preparing to unpack splunkforwarder-9.3.2-d8bb32809498-linux-2.6-amd64.deb ...
Unpacking splunkforwarder (9.3.2) ...
Setting up splunkforwarder (9.3.2) ...
find: '/opt/splunkforwarder/lib/python3.7/site-packages': No such file or directory
find: '/opt/splunkforwarder/lib/python3.9/site-packages': No such file or directory
complete

```

Step 3 : Start the forwarder and enable boot start :

`sudo /opt/splunkforwarder/bin/splunk start --accept-license`

```

(root@kali)-[/tmp]
# /opt/splunkforwarder/bin/splunk start --accept-license-
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
SPLUNK GENERAL TERMS

Splunk> Take the sh out of IT.

Checking prerequisites...
  Checking mgmt port [8089]: open
    Creating: /opt/splunkforwarder/var/lib/splunk
    Creating: /opt/splunkforwarder/var/run/splunk
    Creating: /opt/splunkforwarder/var/run/splunk/appserver/i18n
    Creating: /opt/splunkforwarder/var/run/splunk/appserver/modules/static/css
    Creating: /opt/splunkforwarder/var/run/splunk/upload
    Creating: /opt/splunkforwarder/var/run/splunk/search_telemetry
    Creating: /opt/splunkforwarder/var/run/splunk/search_log
    Creating: /opt/splunkforwarder/var/spool/splunk
    Creating: /opt/splunkforwarder/var/spool/dimmoncache
    Creating: /opt/splunkforwarder/var/lib/splunk/authdb
    Creating: /opt/splunkforwarder/var/lib/splunk/hashdb
    Creating: /opt/splunkforwarder/var/run/splunk/sessions
New certs have been generated in '/opt/splunkforwarder/etc/auth'.
  Checking conf files for problems...
    Done
  Checking default conf files for edits...
    Validating installed files against hashes from '/opt/splunkforwarder/splunkforwarder-9.3.2-d8bb32809498-linux-2.6-x86_64-manifest'
    All installed files intact.
    Done
All preliminary checks passed.

Starting splunk server daemon (splunkd)...
Done

(root@kali)-[/tmp]
#

```

Step 4 : enable boot start :

`sudo /opt/splunkforwarder/bin/splunk enable boot-start --systemd-managed 0`

```

(root@kali)-[/tmp]
# /opt/splunkforwarder/bin/splunk enable boot-start --systemd-managed 0
Init script installed at /etc/init.d/splunk.
Init script is configured to run at boot.

```

```

(root@kali)-[/opt]
# /opt/splunkforwarder/bin/splunk start
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
The splunk daemon (splunkd) is already running.

```

Step 5 : Configure forward-server to point to Splunk Enterprise server IP and receiving port (e.g., 10.0.0.1:9997):

(already added so showing list of active forwards)

```
(trixie@vbox)-[~]
$ sudo /opt/splunkforwarder/bin/splunk list forward-server

Your session is invalid. Please login.
Splunk username: arjun
Password:
Active forwards:
    192.168.1.7:9997
Configured but inactive forwards:
    None
```

Firstly start the booted Splunk forwarder and the add server ip address and default port no and restart the Splunk forwarder.

```
(trixie@vbox)-[~]
$ sudo /opt/splunkforwarder/bin/splunk restart

systemctl: /opt/splunkforwarder/lib/libcrypto.so.3: version 'OPENSSL_3.4.0' not found (required by /usr/lib/x86_64-linux-gnu/systemd/libsystemd-shared-258.so)
systemctl: /opt/splunkforwarder/lib/libcrypto.so.3: version 'OPENSSL_3.4.0' not found (required by /usr/lib/x86_64-linux-gnu/systemd/libsystemd-shared-258.so)
splunkd is not running.
systemctl: /opt/splunkforwarder/lib/libcrypto.so.3: version 'OPENSSL_3.4.0' not found (required by /usr/lib/x86_64-linux-gnu/systemd/libsystemd-shared-258.so)

Splunk> Map. Reduce. Recycle.

Checking prerequisites ...
  Checking mgmt port [8089]: open
New certs have been generated in '/opt/splunkforwarder/etc/auth'.
  Checking conf files for problems ...
    Done
  Checking default conf files for edits ...
    Validating installed files against hashes from '/opt/splunkforwarder/splunkforwarder-10.0.1-c486717c322b-linux-amd64-manifest'
    All installed files intact.
    Done
All preliminary checks passed.

Starting splunk server daemon (splunkd)...
```

Lastly Add Monitor Logs

sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/auth.log.

sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/syslog.

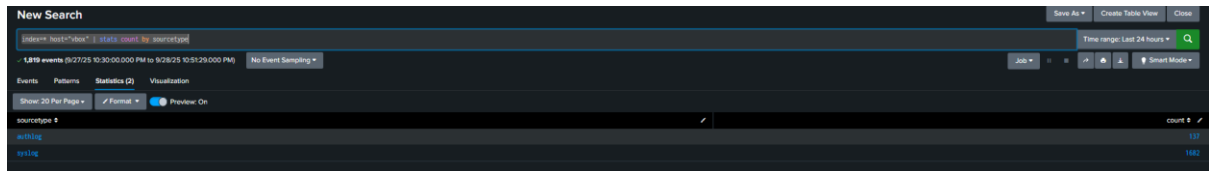
To analyze the logs received from the monitored Linux machine, launch the Splunk Enterprise application and navigate to the "Search & Reporting" app within the Apps section.

Select the desired host to view and analyze the collected log data. This interface allows for comprehensive search, filtering, and visualization of logs originating from the monitored Linux system, enabling effective troubleshooting, security monitoring, and operational insights.

[illegible][illegible]

Given below is a general query that counts the number of events grouped by sourcetype for the host "vbox".

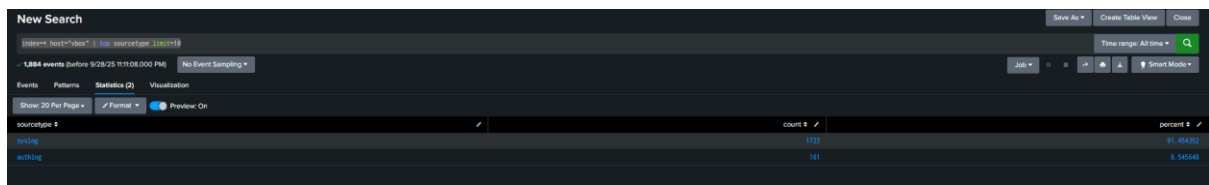
```
index=* host="vbox" | stats count by sourcetype
```



sourcetype	count
syslog	117
winlog	1661

```
index=* host="vbox" | top sourcetype limit=10
```

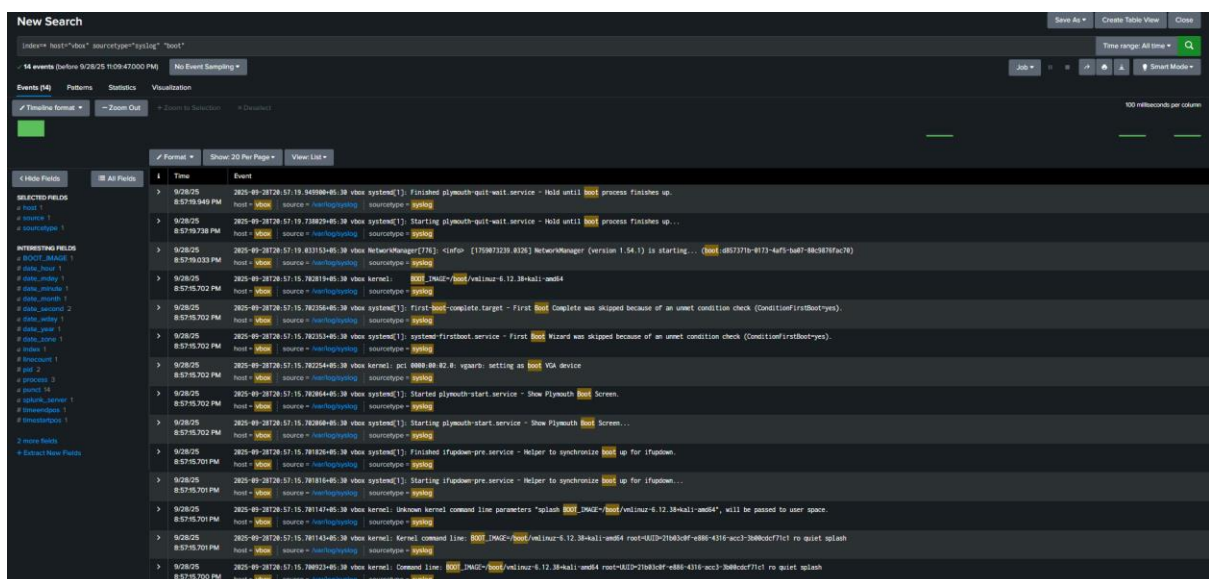
Lists the top 10 most frequent sourcetypes by event count, ordered from highest to lowest count.



sourcetype	count
syslog	1723
winlog	161

```
index=* host="vbox" sourcetype="syslog" "boot"
```

Query used to identify system boot and startup related messages in the logs, which are important for analyzing system restarts, uptime, and related events.



Time	Event
9/28/25 8:57:59.949 PM	2025-09-28T26:57:59.949880-05:30 vbox system[1]: finished plymouth-quit-wait.service - Hold until boot process finishes up.
9/28/25 8:57:59.738 PM	2025-09-28T26:57:59.738203-05:30 vbox system[1]: Starting plymouth-quit-wait.service - Hold until boot process finishes up...
9/28/25 8:57:59.033 PM	2025-09-28T26:57:59.83153-05:30 vbox NetworkManager[778]: nmcli [170873229.8126] NetworkManager (version 1.54.1) is starting... (boot d857371b-9f73-d4f5-ba87-8b38787a678)
9/28/25 8:57:45.702 PM	2025-09-28T26:57:15.782319-05:30 vbox kernel: BOOT_IMAGE=/boot/vmlinuz-6.12.38-hall-uefi4
9/28/25 8:57:45.702 PM	2025-09-28T26:57:15.782304-05:30 vbox system[1]: first- boot -complete.target - First boot Complete was skipped because of an unset condition check (ConditionFirstBoot=yes).
9/28/25 8:57:45.702 PM	2025-09-28T26:57:15.782323-05:30 vbox system[1]: systemd-firstboot.service - First boot Wizard was skipped because of an unset condition check (ConditionFirstBoot=yes).
9/28/25 8:57:45.702 PM	2025-09-28T26:57:15.782254-05:30 vbox kernel: pci 0000:80:02.0: ugarb: setting as boot VGA device
9/28/25 8:57:45.702 PM	2025-09-28T26:57:15.782644-05:30 vbox system[1]: Started plymouth-start.service - Show Plymouth boot Screen.
9/28/25 8:57:45.702 PM	2025-09-28T26:57:15.782648-05:30 vbox system[1]: Starting plymouth-start.service - Show Plymouth boot Screen...
9/28/25 8:57:45.701 PM	2025-09-28T26:57:15.781264-05:30 vbox system[1]: finished ifupdown-pre.service - Helper to synchronize boot up for ifupdown.
9/28/25 8:57:45.701 PM	2025-09-28T26:57:15.781815-05:30 vbox system[1]: Starting ifupdown-pre.service - Helper to synchronize boot up for ifupdown...
9/28/25 8:57:45.701 PM	2025-09-28T26:57:15.78147-05:30 vbox kernel: Unknown kernel command line parameters 'splash BOOT_IMAGE=/boot/vmlinuz-6.12.38-hall-uefi4 ', will be passed to user space.
9/28/25 8:57:45.701 PM	2025-09-28T26:57:15.78143-05:30 vbox kernel: Kernel command line: BOOT_IMAGE=/boot/vmlinuz-6.12.38-hall-uefi4 root=UUID-218b3d8f-e886-4316-ae33-3a8bdc87f1c1 ro quiet splash
9/28/25 8:57:45.700 PM	2025-09-28T26:57:15.780223-05:30 vbox kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.12.38-hall-uefi4 root=UUID-218b3d8f-e886-4316-ae33-3a8bdc87f1c1 ro quiet splash

Conclusion

This project successfully demonstrates the implementation of a centralized log analysis and monitoring solution using Splunk Enterprise. Through configuring Splunk Universal Forwarders on Kali Linux and connecting them to a Splunk Enterprise instance on Windows, log data from diverse sources, including system, authentication, and boot logs, was collected efficiently.

The project covered essential steps such as setting up data ingestion pipelines, defining indexes and sourcetypes, and creating tailored searches and dashboards for real-time security monitoring and operational insights. The ability to analyze logs in a unified platform facilitated quick identification of security incidents, system health issues, and user activities, thereby enhancing situational awareness and response capabilities.

This approach not only streamlines log management across heterogeneous environments but also supports proactive threat hunting, compliance reporting, and continuous improvement of security posture. The knowledge gained from this project can serve as a foundation for expanding log analytics capabilities and integrating additional data sources in future deployments.

References

- Official Splunk Documentation and User Guides
- Kali Linux Official Documentation and Network Configuration Guides
- Online Splunk Community Forums and Knowledge Base Articles
- Technical blogs and tutorials on Splunk Universal Forwarder configuration and log analysis
- Network and system administration textbooks and best practices references

Contributions

- Consolidated hands-on knowledge of deploying and configuring Splunk Enterprise and Universal Forwarder in a heterogeneous environment involving Kali Linux and Windows
- Developed custom log monitoring and forwarding strategies to ensure relevant log data is collected and analyzed effectively
- Constructed meaningful Splunk searches and dashboards to facilitate real-time security monitoring and operational insights
- Documented the end-to-end process of setting up, troubleshooting, and optimizing Splunk log ingestion pipelines
- Provided a foundation for future security monitoring projects by demonstrating practical integration and log analysis workflows

This project reflects the application of acquired knowledge and resources to build a robust log analysis framework using Splunk, contributing real-world value and learning opportunities.

